

# 3-way ANOVA Report

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## Introduction

In this report we are going to write a code that runs a 3-way ANOVA with a  $2^k$  FD for a dataset with 3 parameters A,B,C that affect a certain response in carbonated drinks production.

## The Data

As always we insert the data.

```
#|echo: true
Y<-c(-3,-1,-1,0,-1,0,1,1,0,1,2,1,2,3,6,5) #each couple of data points refer to the same factor
```

We setup the design matrix:

```
A  <- rep(c(-1,1), each = 8)
B  <- rep(rep(c(-1,1), each = 4), 2)
C  <- rep(rep(c(-1,1), each = 2), 4)
AB <- A*B
AC <- A*C
BC <- C*B
ABC <- A*C*B
#We create the design matrix
Design <- data.frame(
  Intercept = 1,
  A  = A,
  B  = B,
  C  = C,
  AB = AB,
  AC = AC,
  BC = BC,
```

```

    ABC = ABC
)
Design

```

|    | Intercept | A  | B  | C  | AB | AC | BC | ABC |
|----|-----------|----|----|----|----|----|----|-----|
| 1  | 1         | -1 | -1 | -1 | 1  | 1  | 1  | -1  |
| 2  | 1         | -1 | -1 | -1 | 1  | 1  | 1  | -1  |
| 3  | 1         | -1 | -1 | 1  | 1  | -1 | -1 | 1   |
| 4  | 1         | -1 | -1 | 1  | 1  | -1 | -1 | 1   |
| 5  | 1         | -1 | 1  | -1 | -1 | 1  | -1 | 1   |
| 6  | 1         | -1 | 1  | -1 | -1 | 1  | -1 | 1   |
| 7  | 1         | -1 | 1  | 1  | -1 | -1 | 1  | -1  |
| 8  | 1         | -1 | 1  | 1  | -1 | -1 | 1  | -1  |
| 9  | 1         | 1  | -1 | -1 | -1 | -1 | 1  | 1   |
| 10 | 1         | 1  | -1 | -1 | -1 | -1 | 1  | 1   |
| 11 | 1         | 1  | -1 | 1  | -1 | 1  | -1 | -1  |
| 12 | 1         | 1  | -1 | 1  | -1 | 1  | -1 | -1  |
| 13 | 1         | 1  | 1  | -1 | 1  | -1 | -1 | -1  |
| 14 | 1         | 1  | 1  | -1 | 1  | -1 | -1 | -1  |
| 15 | 1         | 1  | 1  | 1  | 1  | 1  | 1  | 1   |
| 16 | 1         | 1  | 1  | 1  | 1  | 1  | 1  | 1   |

## The Results

We calculate the estimator:

```

X <- cbind(1, A, B, C , AB, AC, BC , ABC)
beta_hat <- solve(t(X) %*% X) %*% t(X) %*% Y
beta_hat

```

```

      [,1]
      1.000
A      1.500
B      1.125
C      0.875
AB     0.375
AC     0.125
BC     0.250
ABC    0.250

```

We see that the analysis matches the results we expected and the results we derived by doing the calculations by hand. We will now do it with a single line of code to calculate p-values as well.

```
df <- data.frame(Y, A, B , C)
model <- lm(Y ~ A * B* C ,data = df)
summary(model)
```

Call:

```
lm(formula = Y ~ A * B * C, data = df)
```

Residuals:

| Min  | 1Q   | Median | 3Q  | Max |
|------|------|--------|-----|-----|
| -1.0 | -0.5 | 0.0    | 0.5 | 1.0 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t ) |     |
|-------------|----------|------------|---------|----------|-----|
| (Intercept) | 1.0000   | 0.1976     | 5.060   | 0.000977 | *** |
| A           | 1.5000   | 0.1976     | 7.589   | 6.37e-05 | *** |
| B           | 1.1250   | 0.1976     | 5.692   | 0.000459 | *** |
| C           | 0.8750   | 0.1976     | 4.427   | 0.002205 | **  |
| A:B         | 0.3750   | 0.1976     | 1.897   | 0.094350 | .   |
| A:C         | 0.1250   | 0.1976     | 0.632   | 0.544737 |     |
| B:C         | 0.2500   | 0.1976     | 1.265   | 0.241504 |     |
| A:B:C       | 0.2500   | 0.1976     | 1.265   | 0.241504 |     |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

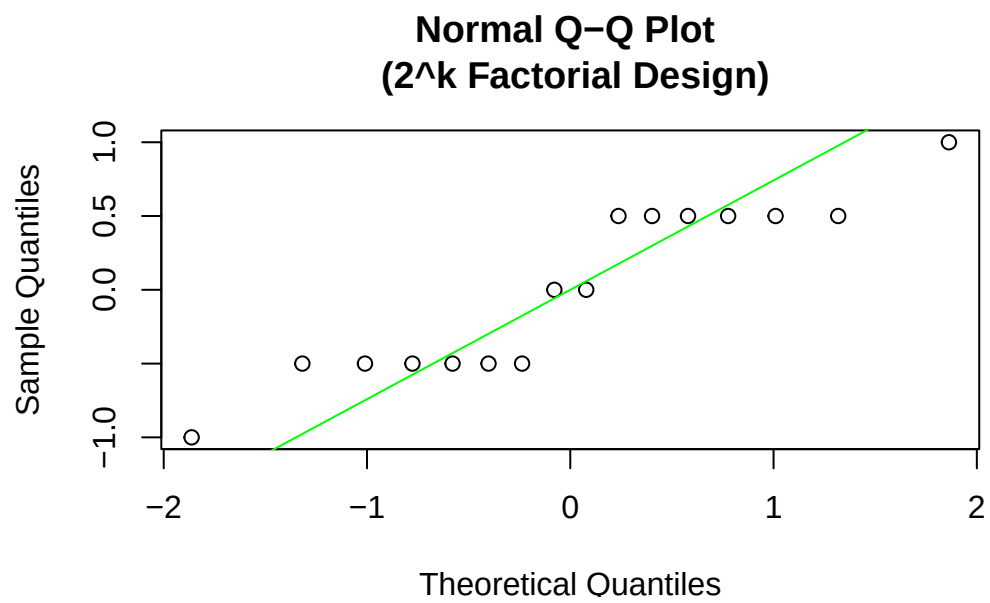
Residual standard error: 0.7906 on 8 degrees of freedom

Multiple R-squared: 0.9359, Adjusted R-squared: 0.8798

F-statistic: 16.69 on 7 and 8 DF, p-value: 0.0003474

Of course no analysis would be complete without a Normal Probability Plot:

```
qqnorm(residuals(model),main = "Normal Q-Q Plot \n (2^k Factorial Design)")
qqline(residuals(model),col="green")
```



It seems like the residuals from lines with a relatively stable angle to our fitted one.